

Patient Management System - Project Report

This report provides a detailed explanation of the Patient Management System project developed using Java, Maven, JDBC, and MySQL. The project was designed to manage patient records, doctor details, appointments, and billing operations in a healthcare environment.

1. Project Overview

The Patient Management System is a console-based Java application developed to simplify hospital and clinic management tasks. The application allows administrators to manage patients, doctors, appointments, and billing information efficiently. The system follows a layered architecture where different responsibilities are separated into modules such as controller, service, DAO, model, and utility layers. Key Functionalities:

- Add, update, delete, and search patient records
 - Manage doctor information
 - Book and view appointments
 - Prevent appointment slot conflicts
 - Generate and manage billing information
 - Store and retrieve records from MySQL database
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2. Technologies Used

Programming Language: Java

Database: MySQL

Database Connectivity: JDBC (Java Database Connectivity)

Build Tool: Maven

IDE Used: Eclipse IDE

Version Control: Git & GitHub

JDK Version: Java 21 (Project initially developed using Java 8)

Architecture: Layered Architecture (Controller → Service → DAO → Database)

3. Roles & Responsibilities

During the development of this project, the following responsibilities were handled:

- Designed the project structure and layered architecture
- Created Java model classes for Patient, Doctor, Appointment, and Billing
- Implemented DAO classes for database operations using JDBC
- Developed service layer logic for validation and business operations
- Managed MySQL database tables and connections
- Implemented appointment conflict checking logic

- Integrated Maven dependencies and project configuration
 - Performed debugging, testing, and error fixing
 - Managed project source code using GitHub
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4. Challenges Faced

Several technical challenges were encountered during the project development:

1. Database Connectivity Issues:

Initially, JDBC connection setup and MySQL driver configuration caused connection failures. This was resolved by properly configuring Maven dependencies and JDBC URLs.

2. Maven and JDK Compatibility:

The project initially faced issues due to Java version mismatch and Maven build configuration problems. These issues were fixed by updating the compiler configuration and build path settings.

3. Package Structure Errors:

Incorrect package hierarchy and duplicate files caused compilation errors. The structure was reorganized into proper controller, service, dao, model, and util packages.

4. Appointment Validation:

Implementing appointment conflict checking logic required validation to prevent duplicate doctor bookings for the same time slot.

5. Project Outcome

The project was successfully completed and achieved the following outcomes:

- Successfully implemented CRUD operations for hospital management
- Integrated Java application with MySQL database
- Improved understanding of layered architecture and JDBC
- Learned Maven project management and dependency handling
- Developed practical experience in debugging and troubleshooting Java applications
- Gained knowledge of real-world software development workflow using GitHub

The project demonstrates strong understanding of Java backend development concepts and database integration.

Conclusion:

The Patient Management System project helped in understanding core software engineering concepts such as modular design, database integration, layered architecture, and project management using Maven and GitHub. The project can be further enhanced in the future by adding GUI support, authentication, REST APIs, and web-based deployment.